

Sequences, Series & Complex Numbers

ANSWER KEY



Arithmetic sequences

- Find the 12th term of the arithmetic sequence 5, 9, 13, ...
 $a_1 = 5, d = 4. a_{12} = 5 + (11)(4) = 49.$
- The 4th term of an arithmetic sequence is 18, and the 9th term is 38. Find the first term and the common difference.
 $a_1 + 3d = 18$ and $a_1 + 8d = 38$. Subtracting: $5d = 20 \Rightarrow d = 4$, so $a_1 = 18 - 12 = 6.$
- Find the sum of the first 20 terms of the arithmetic sequence 7, 11, 15, ...
 $a_1 = 7, d = 4, a_{20} = 7 + (19)(4) = 83. S_{20} = \frac{20}{2}(7 + 83) = 10(90) = 900.$
- Find the number of terms in the arithmetic sequence 6, 10, 14, ..., 150.
 $a_1 = 6, d = 4. 6 + 4(n - 1) = 150 \Rightarrow 4(n - 1) = 144 \Rightarrow n - 1 = 36 \Rightarrow n = 37.$

Geometric sequences

- Find the 7th term of the geometric sequence 3, 6, 12, ...
 $a_1 = 3, r = 2. a_7 = 3(2^6) = 3(64) = 192.$
- In a geometric sequence with all positive terms, the 3rd term is 20 and the 5th term is 80. Find r and a_1 .
 $a_1 r^2 = 20$ and $a_1 r^4 = 80$. Dividing: $r^2 = 4$, and since all terms are positive, $r = 2$. Then $a_1 = \frac{20}{4} = 5.$
- Given that $x - 1$, $x + 1$, and $3x - 1$ are consecutive terms of a geometric sequence, find the possible value(s) of x .
 $(x + 1)^2 = (x - 1)(3x - 1) \Rightarrow x^2 + 2x + 1 = 3x^2 - 4x + 1 \Rightarrow 0 = 2x^2 - 6x = 2x(x - 3)$. So $x = 0$ or $x = 3$. If $x = 0$, the terms are $-1, 1, -1$ (ratio -1 , valid). If $x = 3$, the terms are $2, 4, 8$ (ratio 2 , valid).
- Find the sum of the first 6 terms of the geometric sequence 5, 15, 45, ...
 $a_1 = 5, r = 3. S_6 = \frac{5(3^6 - 1)}{3 - 1} = \frac{5(728)}{2} = 1820.$

Series and sigma notation

- Evaluate $\sum_{k=1}^5 (3k - 2)$.
Terms: $1, 4, 7, 10, 13$. Sum = $1 + 4 + 7 + 10 + 13 = 35.$
- Write the series $3 + 7 + 11 + \dots + 39$ in sigma notation, then evaluate it.
The general term is $a_k = 4k - 1$ for $k = 1, \dots, 10$ (since $4(10) - 1 = 39$). $\sum_{k=1}^{10} (4k - 1) = 210.$

- 11 Evaluate $\sum_{n=1}^6 2^n$.
 $2 + 4 + 8 + 16 + 32 + 64 = 126$.

Geometric series

- 12 Find the sum of the first 8 terms of the geometric sequence 2, 6, 18, ...
 $a_1 = 2, r = 3. S_8 = \frac{2(3^8 - 1)}{3 - 1} = \frac{2(6561 - 1)}{2} = 6560$.
- 13 Find the sum to infinity of $16 - 8 + 4 - 2 + \dots$
 $a_1 = 16, r = -\frac{1}{2}. S_\infty = \frac{16}{1 - (-\frac{1}{2})} = \frac{16}{3/2} = \frac{32}{3}$.
- 14 Express the repeating decimal $0.4\overline{5}$ (that is, $0.45555\dots$) as a geometric series and find its exact fractional value.
 $0.4\overline{5} = 0.4 + (0.05 + 0.005 + 0.0005 + \dots)$. The bracketed part is geometric with $a_1 = 0.05, r = 0.1$, summing to $\frac{0.05}{0.9} = \frac{1}{18}$. Total: $\frac{2}{5} + \frac{1}{18} = \frac{36}{90} + \frac{5}{90} = \frac{41}{90}$.

Complex number arithmetic

- 15 Given $z_1 = 5 - 2i$ and $z_2 = -3 + 4i$, find $z_1 + z_2$ and $z_1 - z_2$.
 $z_1 + z_2 = (5 - 3) + (-2 + 4)i = 2 + 2i. z_1 - z_2 = (5 - (-3)) + (-2 - 4)i = 8 - 6i$.
- 16 Multiply $(3 + 2i)(4 - 5i)$.
 $12 - 15i + 8i - 10i^2 = 12 - 7i + 10 = 22 - 7i$.
- 17 Simplify i^{23} , using $i^2 = -1$.
Powers of i cycle with period 4: $i^1 = i, i^2 = -1, i^3 = -i, i^4 = 1$. Since $23 = 4(5) + 3, i^{23} = i^3 = -i$.
- 18 Simplify $(2 - 3i) - (5 + i) + (1 - 4i)$.
 $(2 - 5 + 1) + (-3 - 1 - 4)i = -2 - 8i$.

Complex conjugates and division

- 19 Find the conjugate of $z = 6 + 7i$, and compute $z\bar{z}$.
 $\bar{z} = 6 - 7i. z\bar{z} = 6^2 + 7^2 = 36 + 49 = 85$.
- 20 Simplify $\frac{4 + 3i}{1 - 2i}$ by multiplying by the conjugate of the denominator.
 $\frac{4 + 3i}{1 - 2i} \cdot \frac{1 + 2i}{1 + 2i} = \frac{(4 + 3i)(1 + 2i)}{1^2 + 2^2} = \frac{4 + 8i + 3i + 6i^2}{5} = \frac{-2 + 11i}{5} = -\frac{2}{5} + \frac{11}{5}i$.
- 21 Simplify $\frac{5}{3 + i}$ by multiplying by the conjugate of the denominator.
 $\frac{5}{3 + i} \cdot \frac{3 - i}{3 - i} = \frac{5(3 - i)}{9 + 1} = \frac{15 - 5i}{10} = \frac{3}{2} - \frac{1}{2}i$.

Modulus of complex numbers

- 22 Find $|z|$ for $z = -8 + 15i$.

$$|z| = \sqrt{(-8)^2 + 15^2} = \sqrt{64 + 225} = \sqrt{289} = 17.$$

- 23 If $z_1 = 3 + 4i$ and $z_2 = 1 - i$, find $|z_1 z_2|$.

$$z_1 z_2 = (3 + 4i)(1 - i) = 3 - 3i + 4i - 4i^2 = 7 + i, \text{ so } |z_1 z_2| = \sqrt{49 + 1} = \sqrt{50} = 5\sqrt{2}. \text{ (This matches } |z_1||z_2| = 5 \cdot \sqrt{2}.)$$

- 24 If $z = 4 - 3i$, find $|z|$ and $|2z|$.

$$|z| = \sqrt{16 + 9} = \sqrt{25} = 5. \quad |2z| = 2|z| = 10.$$

Quadratic equations with complex roots

- 25 Solve $x^2 + 4x + 13 = 0$, giving your answer in the form $a + bi$.

$$\text{By the quadratic formula: } x = \frac{-4 \pm \sqrt{16 - 52}}{2} = \frac{-4 \pm \sqrt{-36}}{2} = \frac{-4 \pm 6i}{2} = -2 \pm 3i.$$

- 26 One root of a quadratic equation with real coefficients is $2 - 5i$. Find the other root, and write the equation in standard form.

Since the coefficients are real, the other root is $2 + 5i$. Sum of roots = 4; product = $(2 - 5i)(2 + 5i) = 4 + 25 = 29$. Equation: $x^2 - 4x + 29 = 0$.

- 27 Solve $2x^2 - 4x + 5 = 0$, giving your answer in the form $a + bi$.

$$x = \frac{4 \pm \sqrt{16 - 40}}{4} = \frac{4 \pm \sqrt{-24}}{4} = \frac{4 \pm 2\sqrt{6}i}{4} = 1 \pm \frac{\sqrt{6}}{2}i.$$

A well-designed word problem: a bouncing ball

- 28 This question has three parts. A ball is dropped from a height of 20 meters. Each time it bounces, it rises to 60% of its previous height. (a) Find the height of the ball's rebound after its 4th bounce. (b) Write an expression for the total vertical distance traveled by the ball as it bounces infinitely many times, and evaluate it. (c) After which bounce does the rebound height first fall below 1 meter?

(a) The rebound height after the n th bounce is $h_n = 20(0.6)^n$. $h_4 = 20(0.6)^4 = 20(0.1296) = 2.592$ meters. (b) The ball falls 20 m initially, then for each bounce it rises and falls the same rebound height. Total distance = $20 + 2 \sum 20(0.6)^n = 20 + 2\left(\frac{12}{1 - 0.6}\right) = 20 + 2(30) = 80$ meters. (c) We need $20(0.6)^n < 1 \Rightarrow (0.6)^n < 0.05$. Checking: $(0.6)^5 = 0.0778$ gives $h_5 = 1.555$ m (still above 1 m), while $(0.6)^6 = 0.0467$ gives $h_6 = 0.933$ m (below 1 m). The rebound height first falls below 1 meter after the 6th bounce.